SQL PORTFOLIO PROJECT

Inventory Lot Expiration & Aging Analysis

Inventory | Lot Tracking | Aging Buckets | CTEs | Warehouse Compliance

Project Summary

This query identifies inventory lots approaching or past expiration and segments them into six aging buckets (Current, 1-30, 31-60, 61-90, 91-120, 120+ days). It surfaces license plate and lot number traceability data to support FIFO compliance, quality holds, and proactive disposition decisions.

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1 PROJECT OVERVIEW

Platform:	Oracle ERP -- ORAMART (SIM Daily Inventory operational view)
Query Type:	Multi-CTE aging bucket analysis with LEFT JOIN fan-out
Grain:	Lot Number / Item
Key Outputs:	Expiration date, days overdue, six aging buckets (Current to 120+)
Filter:	lot_status IN ('GOOD','Active','MCRO') -- excludes held/quarantine lots
Audience:	Warehouse Management, Quality Assurance, Inventory Control

2 BUSINESS PROBLEM

Warehouse and quality teams needed a daily snapshot of which inventory lots were expiring or already past their expiration date -- categorized into actionable aging buckets. Without this report, teams relied on manual spreadsheets updated weekly, which missed short-dated lots and created food safety compliance risk.

- Lots status-filtered to only active/good lots -- quarantined or held lots excluded automatically.
- Aging buckets enable tiered responses: Current = monitor, 1-30 = prioritize pick, 31-60 = quality review, 90+ = disposition decision.
- License plate and lot number together provide full warehouse traceability for recalls or audits.
- SYSDATE comparison means the report is always current -- no manual date parameter needed.
- NVL(0) ensures every item/lot row appears even if it falls in only one bucket.

3 TECHNICAL EXPLANATION

- WOR_DaysDue -- base CTE: computes DaysOverDue = TRUNC(LOT_EXPIRATION_DATE - SYSDATE) for all active lots. Negative = expired; positive = days remaining.
- WORCurrent -- lots where DaysOverDue <= 0 (expired or expiring today); ABS() converts negatives to positive for display.
- WOR30Days through WOROver120Days -- each CTE filters to its specific range using numeric comparisons on DaysOverDue.
- Final SELECT -- LEFT JOINs all bucket CTEs back to the base on item_no. NVL(0) ensures items not in a bucket show 0 rather than NULL.
- ORDER BY item_no, DaysOverDue -- groups all lots for an item together, soonest-expiring first.

Key Techniques: CTE chaining, aging bucket pattern, SYSDATE arithmetic, TRUNC date math, ABS(), NVL(), LEFT JOIN fan-out

4 FULL SQL QUERY -- COMPLETE AND UNABRIDGED

```
WITH WOR_DaysDue AS (
  SELECT
    t0.item_no,
    t0.LOT_EXPIRATION_DATE,
    TRUNC(t0.LOT_EXPIRATION_DATE - SYSDATE) AS DaysOverDue,
    t0.lot_number
  FROM ORAMART.sim_daily_inventory t0
  WHERE t0.lot_status IN ('GOOD','Active','MCRO')
),
WORCurrent AS (
  SELECT t0.lot_number, t0.item_no,
         ABS(t0.DaysOverDue) AS DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue <= 0
),
WOR30Days AS (
  SELECT t0.lot_number, t0.item_no, t0.DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue > 0 AND t0.DaysOverDue <= 30
),
WOR60Days AS (
  SELECT t0.lot_number, t0.item_no, t0.DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue > 30 AND t0.DaysOverDue <= 60
),
WOR90Days AS (
  SELECT t0.lot_number, t0.item_no, t0.DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue > 60 AND t0.DaysOverDue <= 90
),
WOR120Days AS (
  SELECT t0.lot_number, t0.item_no, t0.DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue > 90 AND t0.DaysOverDue <= 120
),
WOROver120Days AS (
  SELECT t0.lot_number, t0.item_no, t0.DaysOverDue
  FROM WOR_DaysDue t0 WHERE t0.DaysOverDue > 120
)
SELECT
  t1.lot_number,
  t1.item_no,
  NVL(t2.DaysOverDue, 0) AS "Current",
  NVL(t3.DaysOverDue, 0) AS "1 - 30 Days",
  NVL(t4.DaysOverDue, 0) AS "31 - 60 Days",
  NVL(t5.DaysOverDue, 0) AS "61 - 90 Days",
  NVL(t6.DaysOverDue, 0) AS "91 - 120 Days",
  NVL(t7.DaysOverDue, 0) AS "Over 120 Days"
FROM WOR_DaysDue T1
LEFT JOIN WORCurrent     t2 ON T1.item_no = t2.item_no
LEFT JOIN WOR30Days      t3 ON T1.item_no = t3.item_no
LEFT JOIN WOR60Days      t4 ON T1.item_no = t4.item_no
LEFT JOIN WOR90Days      t5 ON T1.item_no = t5.item_no
LEFT JOIN WOR120Days     t6 ON T1.item_no = t6.item_no
LEFT JOIN WOROver120Days t7 ON T1.item_no = t7.item_no
ORDER BY t1.item_no, t1.DaysOverDue
```

Lot #	Item #	Current	1-30 Days	31-60 Days	61-90 Days	91-120 Days	120+ Days
LOT-4421	ITEM-050	3	0	0	0	0	0
LOT-4422	ITEM-050	0	18	0	0	0	0
LOT-4423	ITEM-051	0	0	45	0	0	0
LOT-4424	ITEM-052	0	0	0	0	105	0
LOT-4425	ITEM-053	0	0	0	0	0	142

** Lot numbers and item numbers masked. Bucket value = days in that range (0 if not applicable).*